

A Frequency-Resolved Gaussian Channel Bound for MQM Thermal Capture

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Abstract—We refine the thermal-to-vacuum comparison for the fixed MQM flat-band capture reference at inverse temperature $\beta = 5/16$ and deadline $t = 16$. Gaussian pairing of a bounded displacement drive yields an all-orders channel bound involving the absolute real and imaginary covariance integrals. A closed analytic kernel envelope and exact rational majorants certify total retained error below 2^{-13} , improving the earlier variance-only allowance without changing the spectrum, temperature, or deadline. The requested 2^{-16} target remains open. Signed scalar filters are diagnostic and cannot replace a noncommuting channel calculation. The result is conditional on the declared Gaussian continuum reference; it is not a thermal trajectory simulation, a physical latch realization, permanent retention, or independent external review.

Index Terms—Quantum control, thermal baths, Gaussian noise, channel bounds, non-Markovian dynamics.

I. FROZEN SOURCE AND RETAINED DOMAIN

The Mayes Quantum Method (MQM) reference has compact spectrum $J(\omega) = \kappa/(2\pi)$ on $[\Omega - B, \Omega + B]$, with $\Omega = 64$, $B = 16$, $\kappa = 8$, coherent port couplings $g = h = 1$, and $\lambda^2 = 4\sqrt{2}$. We retain $\beta = 5/16$, $t = 16$, and target 2^{-16} . The source launch is the admitted port state $pR0$, with an arbitrary identity bank and reference. Thermal dynamics uses the full eight-dimensional admitted clock-auxiliary sector, not the four live vacuum states. The original reservoir is traced in the retained channel.

The inherited vacuum ceiling is

$$E_0 = \frac{3}{262144} + 2^{-128}. \quad (1)$$

We assume the parent Gaussian continuum limit of finite-mode reduced dynamics, with stationary free bath and factorized launch. Material realization, native admission, complete-word compilation, and reservoir refresh remain separate obligations.

Finite-mode thermal coherent-state mixtures and bath displacement express the reduced channel as vacuum-reference dynamics averaged over a centered Gaussian drive. This is a proof representation, not scheduled control. Gaussian environmental decomposition is prior art [1].

II. ROTATING FRAME AND CORRELATION KERNEL

Conservation of total excitation permits removal of the common carrier Ω . The bath frequencies become detunings $\delta \in [-B, B]$. With

$$n_\beta(\omega) = \frac{1}{e^{\beta\omega} - 1}, \quad (2)$$

the displacement field ζ is circular Gaussian with

$$C(s) = \mathbb{E}[\zeta(s)\zeta(0)^*] \quad (3)$$

$$= \int_{-B}^B J(\Omega + \delta) n_\beta(\Omega + \delta) e^{-i\delta s} d\delta, \quad (4)$$

$$V = C(0). \quad (5)$$

The previous bound was

$$\frac{1}{2} \|\Phi_t^\beta - \Phi_t^0\|_\diamond \leq \min\{1, t^2 V\}. \quad (6)$$

The norm includes arbitrary reference extensions; fixed-launch task error follows by composition and partial trace. In the source system $S = I_C \otimes \sigma_R^-$, so the drive is $\zeta S^\dagger + \zeta^* S$. Write $\zeta = a + ib$. Its two Hermitian quadrature operators have norm one. In the vacuum interaction picture they remain bounded, though they need not commute at different times.

The real covariance matrix has entries $\Re C/2$, $\Im C/2$, $-\Im C/2$, and $\Re C/2$, up to the chosen quadrature sign convention. Its entrywise absolute sum is $|\Re C| + |\Im C|$. Define

$$I_{\text{abs}}(t) = \int_0^t (t-s)(|\Re C(s)| + |\Im C(s)|) ds. \quad (7)$$

III. ALL-ORDERS GAUSSIAN PAIRING BOUND

Proposition 1. Under the declared Gaussian reference,

$$\frac{1}{2} \|\Phi_t^\beta - \Phi_t^0\|_\diamond \leq \min \left\{ 1, t^2 V, \frac{e^{4I_{\text{abs}}(t)} - 1}{2} \right\}. \quad (8)$$

Proof. Expand the vacuum-interaction-picture drive super-operator in its time-ordered Dyson series. Each quadrature commutator has completely bounded trace norm at most two. Odd Gaussian moments vanish. Wick pairing at order $2n$ replaces noise moments by products of pair covariances; taking absolute values is necessary before discarding the noncommuting operator products. Summing quadrature indices gives the nonnegative pair kernel

$$K(s, u) = 4(|\Re C(s-u)| + |\Im C(s-u)|). \quad (9)$$

The sum over all pairings is symmetric in all $2n$ times. Therefore the ordered simplex integral equals $1/(2n)!$ times its full-cube integral. There are $(2n)!/(2^n n!)$ pairings, and

$$\int_{[0,t]^2} K(s, u) ds du = 8I_{\text{abs}}(t). \quad (10)$$

Consequently the order- $2n$ diamond-norm contribution is bounded by $(4I_{\text{abs}})^n/n!$. The resulting exponential majorant guarantees absolute convergence of the averaged series. Sum

$n \geq 1$, divide by two, and use contraction under bath trace. The finite-mode argument passes to the assumed reduced Gaussian continuum limit. This proof uses no Born–Markov approximation or first-photon truncation. $\square$

IV. CLOSED KERNEL ENVELOPE AND EXACT CERTIFICATE

The positive Bose expansion gives the exact kernel series

$$C(s) = \frac{\kappa}{2\pi} \sum_{\ell \geq 1} \frac{e^{-\ell\beta(\Omega-B)} e^{iBs} - e^{-\ell\beta(\Omega+B)} e^{-iBs}}{\ell\beta + is}. \quad (11)$$

Let $r_{\pm} = e^{-\beta(\Omega \pm B)}$ and

$$A = \frac{\kappa}{2\pi} \left(\frac{r_-}{1-r_-} + \frac{r_+}{1-r_+} \right). \quad (12)$$

Triangle inequality and $\ell\beta \geq \beta$ yield $|C(s)| \leq A/\sqrt{\beta^2 + s^2}$. Hence

$$I_{\text{abs}}(t) \leq \sqrt{2} A F(t, \beta), \quad (13)$$

$$F(t, \beta) = t \operatorname{arsinh}(t/\beta) - \sqrt{t^2 + \beta^2} + \beta. \quad (14)$$

This retains correlation-time decay while foregoing signed phase cancellation. It controls all Gaussian orders.

The following rational majorants are independently executable. Positive Taylor lower sums through degree 80 establish $e^{15} > 2^{21}$, $e^{25} > 2^{36}$, and $e^{14/3} > 517/5$. Use $\pi > 3$, $\sqrt{2} < 17/12$, and

$$\frac{t + \sqrt{t^2 + \beta^2}}{\beta} < \frac{2t + \beta}{\beta} = \frac{517}{5}. \quad (15)$$

Then

$$A < \frac{4}{3} \left(\frac{1}{2^{21} - 1} + \frac{1}{2^{36} - 1} \right), \quad (16)$$

$$F(16, 5/16) < 16 \frac{14}{3} - 16 + \frac{5}{16} = \frac{2831}{48}. \quad (17)$$

An exponent ceiling is

$$x = \frac{12433696189517}{58512905602490430} < 1. \quad (18)$$

Since $e^x - 1 \leq x/(1-x)$,

$$E_{\beta} < E_0 + \frac{x}{2(1-x)} < \frac{1}{8192} = 2^{-13}. \quad (19)$$

The intermediate thermal debit is exactly bounded by

$$\frac{12433696189517}{117000943812601826}. \quad (20)$$

The total ceiling is approximately 1.17714×10^{-4} . This is a conservative upper certificate, not measured or simulated thermal error. It does not certify the original 2^{-16} target, and it does not prove that the physical error exceeds that target.

V. NUMERICAL DIAGNOSTICS AND CANCELLATION BOUNDARY

The frozen diagnostic uses four Bose terms. The uniform omitted-kernel tail is bounded by

$$\frac{\kappa B}{\pi} \frac{e^{-5\beta(\Omega-B)}}{1 - e^{-\beta(\Omega-B)}}. \quad (21)$$

Midpoint meshes $N = 65536, 131072$ use 80-digit arithmetic. The Lipschitz bound $(2 + 2Bt)V$ gives quadrature radius $(2 + 2Bt)Vt^2/(4N)$, plus t^2 times the Bose tail. Arithmetic is not directed; the exact certificate does not depend on these diagnostics. Adaptive quadrature initially emitted a roundoff warning. Resolving 161 absolute-value cusps removes it; both records are preserved. The refined diagnostic gives $I_{\text{abs}} \approx 2.89478 \times 10^{-5}$ and a thermal debit estimate 5.78989×10^{-5} . Neither is a directed channel certificate. The closed analytic envelope, evaluated with directed intervals, gives total ceiling 7.57572×10^{-5} .

A tempting signed scalar filter is

$$F_2(t) = \int_{-B}^B J_{n\beta} \frac{1 - \cos(\delta t)}{\delta^2} d\delta, \quad (22)$$

with removable value $t^2/2$ at $\delta = 0$. It equals $\int_0^t (t-s) \Re C(s) ds$. The identity is exact for the scalar kernel. Substituting it directly for I_{abs} would discard interaction operators that evolve and do not generally commute. A small F_2 is therefore a diagnostic of possible frequency selectivity, not a complete-channel certificate.

VI. SECOND-ORDER RESPONSE GATE

The same proof bounds all even orders four and higher by

$$\frac{1}{2} \sum_{n \geq 2} \frac{(4I_{\text{abs}})^n}{n!} \leq \frac{x^2}{2(1-x)} < 2^{-25}. \quad (23)$$

Let Δ_2 be the mean ordered product of two interaction-picture drive commutators, composed with the fixed launch and original-bath trace. It is a response correction, not a completely positive channel by itself. A sufficient remaining criterion is

$$\frac{1}{2} \|\Delta_2\|_{\text{tr}} < \frac{127}{2^{25}} - 2^{-128}. \quad (24)$$

Here the trace norm refers to the resulting fixed-launch retained-state correction; the arbitrary identity bank/reference factors through unchanged. With the inherited vacuum ceiling and the higher-order allowance, this criterion would certify 2^{-16} . The full response has not been computed. The scalar F_2 cannot certify it.

Next: certify the full second-order response and a multi-photon truncation reference. Parameters are unchanged; the majorant is an analytical supplement. Permanent hold remains open; physical promotion is zero.

REFERENCES

- [1] S. Luo, N. Lambert, P. Liang, and M. Cirio, “Quantum-classical decomposition of Gaussian quantum environments: A stochastic pseudomode model,” *PRX Quantum*, vol. 4, p. 030316, 2023, doi:10.1103/PRXQuantum.4.030316.